

Reviewer Pack — Minimal Order of Algebraic Topology in Monomial Ideal Comple...

Entry 40858724cef3 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Algebraic Topology in Monomial Ideal Complexity

Entry ID: 40858724cef3 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 20:02:51 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology **Field B** (complexity object): Monomial Ideal Complexity

Statement:

For every boolean function φ with at most n variables, the minimal order of an algebraic topological invariant computed on the van Kampen diagram of its associated monomial ideal $I(\varphi)$ is $O(n^{(3/2)})$. Equivalently, the complexity of computing the invariant is $\Theta(n^{(3/2)})$.

Rationale (proposer's reasoning):

Algebraic topology provides a rich set of invariants that capture topological properties of spaces. By linking these to monomial ideals associated with boolean functions, we may uncover new insights into the complexity of their corresponding circuit representations. The conjecture suggests that the algebraic topological invariant's order correlates with the complexity of computing the ideal, which could lead to a novel understanding of circuit complexity.

Taxonomy category: MONOMIAL_IDEAL_COMPLEXITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 41546eb93d858d47

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal order of an algebraic topological invariant on the van Kampen diagram of monomial ideal $I(\varphi)$ is $O(n^{(3/2)})$, where $n \leq 40$ and the computational complexity is $\Theta(n^{(3/2)})$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.80	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'algebraic topology' AND 'monomial ideal complexity' AND 'invariant computation' - 'van Kampen diagram' AND 'monomial ideal' AND 'algorithmic complexity' - 'topological invariants' IN monomial ideals 'computational complexity'

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return ''.join(random.choice('01') for _ in range(2**n))

    def monomial_ideal_complexity(phi):
        # Placeholder function to simulate complexity calculation
        # Replace with actual implementation if available
        return len(phi) ** 3 / 2

    n = random.randint(5, 40)
    phi = generate_boolean_function(n)
    complexity = monomial_ideal_complexity(phi)

    return {
        "metric_name": "complexity",
        "metric_value": complexity,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": True if complexity <= n ** (3 / 2) else False,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [random.getrandbits(32) for _ in range(30)]
```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(res["metric_value"] for res in results) / len(results)
std_value = math.sqrt(sum((res["metric_value"] - mean_value) ** 2 for res in results) / len(results))
support_fraction = sum(1 for res in results if res["conjecture_holds"]) / len(results)

if all(res["conjecture_holds"] for res in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
else:
    first_failing_seed = next((res["seed"] for res in results if not res["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means the pre-registered support condition could not be unambiguously met. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14180
2	preregistration	ollama_remote	glm4:latest	0	0	10056
3	novelty	ollama_remote	glm4:latest	0	0	8507
4	novelty	ollama_remote	glm4:latest	0	0	8868
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11145
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8224
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8066
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	5645
9	judge	ollama_remote	glm4:latest	0	0	36249

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 110940 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/40858724cef3.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/40858724cef3.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/40858724cef3.tar.gz (if generated)